

附录1: 多种爱因斯坦方程的解:

1. Reissner - Nordstrom 解 带电球对称

电磁真空: 仅有电磁场, 无物质场: $T_{ab} = \frac{1}{4\pi} (F_{ac}F_b^c - \frac{1}{4}g_{ab}F_{cd}F^{cd}) \rightarrow T = T_{ab}g^{ab} = \frac{1}{4\pi} (F_{ac}F^{ac} - F_{cd}F^{cd}) = 0$

electrovacuum

无源2-形式场

$$G_{ab} = R_{ab} - \frac{1}{2}Rg_{ab} = 8\pi T_{ab} \quad F \text{ 有: } \nabla^a F_{ab} = 0, \nabla_{[a}F_{bc]} = 0$$

$$R - 2R = 0 \rightarrow R = 0 \rightarrow R_{ab} = 8\pi T_{ab}$$

爱因斯坦-麦克斯韦方程组

复张量场 $\Sigma_{ab} := F_{ab} + i^*F_{ab}$ ← 2-形式张量的形式

$$\Sigma_{ab}\Sigma^{ab} = F_{ab}F^{ab} - F_{ab}^*F^{ab} + iF_{ab}F^{ab} + iF_{ab}^*F^{ab} = F_{ab}F^{ab} - \frac{F_{cd}F_{cd}\epsilon^{abcd}\epsilon^{abcd}}{2!} + iF_{ab}F^{ab} + iF_{ab}^*\epsilon^{abcd}F_{cd}$$

$$= 2(F_{ab}F^{ab} + iF_{ab}^*F^{ab})$$

类光电磁场: $\Sigma_{ab}\Sigma^{ab} = 0$ ($F_{ab}F^{ab} = 0, F_{ab}^*F^{ab} = 0$)

$$E_a := F_{ab}Z^b; B_a := -F_{ab}^*Z^b \quad \leftarrow B, E \text{ 都可用 } Z \text{ 表出}$$

$$\text{类光: } B^2 = E^2; \vec{E} \cdot \vec{B} = 0$$

求解: 1 对称性: 同史瓦西解, $SO(2)$ 对称:

$$ds^2 = -e^{2\alpha(r)}dt^2 + e^{2\beta(r)}dr^2 + d\hat{s}^2$$

电磁势对称性: A_μ 仅与 r 有关: $A_2 = A_3 = 0$, 规范变换 $\tilde{A}_a = A_a + \nabla_a\chi(r)$ ($F_{ab} = 2\nabla_{[a}A_{b]}$)

$$\tilde{A}_1 = (\partial/\partial r)^a \cdot (A_a + \nabla_a\chi) = A_1 + \frac{\partial\chi}{\partial r} = 0 \rightarrow \text{取 } A_0 \neq 0$$

$$F_{01} = -F_{10} = \frac{dA_0}{dr}, \text{ else } F_{\mu\nu} = 0$$

$$\text{麦方程: } 1) F^{\mu\nu}{}_{;\mu} = \frac{1}{\sqrt{-g}} \frac{\partial}{\partial x^\mu} (\sqrt{-g} F^{\mu\nu}) + \Gamma^\nu_{\alpha\mu} F^{\alpha\mu} \rightarrow -\sqrt{-g} F^{10} = \sqrt{-g} F^{01} = r^2 F_{10} e^{(\alpha+\beta)} \sin\theta$$

$$2) F_{[\mu\nu];\omega] = 0 \quad (\nabla_{[\alpha}\nabla_{\beta}A_{\gamma]} = 0)$$

$$\frac{d}{dr} (r^2 F_{10} e^{(\alpha+\beta)} \sin\theta) = 0 \rightarrow F_{10} = \frac{Q}{r^2} e^{2\beta}$$

$$F_{ab} = -\frac{Q}{r^2} e^{2\beta} \cdot (dt)_a \wedge (dr)_b$$

$$T_{00} = F_{10}^2 \cdot \frac{e^{-2\beta}}{8\pi} \quad T_{11} = -F_{10}^2 \cdot \frac{e^{-2\alpha}}{8\pi} \quad T_{22} = F_{10}^2 \cdot \frac{r^2}{8\pi} e^{-2\alpha-2\beta} \quad T_{33} = \sin^2\theta \cdot T_{22}$$

2. 代入方程 R_{ab} 用度规表出:

$$\text{同史瓦西解: } \begin{cases} R_{00} = e^{-2(\alpha+\beta)} (-\alpha'' + \alpha'\beta' - \alpha'^2 - 2\frac{\alpha'}{r}) \\ R_{11} = e^{2\beta-2\alpha} \cdot R_{00} \end{cases}$$

联立 $R_{ab} = T_{ab}$: $\alpha = -\beta$

$$R_{11} \text{ 与 } R_{22}: (re^{2\alpha})' = 1 - Q^2/r^2 \rightarrow e^{2\alpha} = 1 - \frac{Q^2}{r^2} + \frac{C}{r} \quad e^{2\beta} = (1 - \frac{Q^2}{r^2} + \frac{C}{r})^{-1}$$

解毕: $F_{10} = \frac{Q}{r^2}$, $g_{\mu\nu} = \begin{bmatrix} (1 - \frac{2M}{r} - \frac{Q^2}{r^2}) & & \\ & (1 + \frac{Q^2}{r^2}) & \\ & & \ddots \end{bmatrix}$ $ds^2 = -(1 - \frac{2M}{r} + \frac{Q^2}{r^2}) dt^2 + (1 - \frac{2M}{r} + \frac{Q^2}{r^2})^{-1} dr^2 + d\Omega^2$
 $A_0 = -\frac{Q}{r} \rightarrow Q$ 为电荷 R-N 线元

对静态观者, $Z^a = \sqrt{-g_{00}} (\partial/\partial t)^a$. 取 $(e^1)_a = \sqrt{g_{11}} (dr)_a$, $(e^2)_a = r (d\theta)_a$, $(e^3)_a = r \sin\theta (d\varphi)_a$

$$E_a = F_{ab} Z^b = \frac{Q}{r^2} (e_1)_a, \quad B_a \equiv {}^*F_{ab} Z^b = 0$$

1) $F_{ab} F^{ab} \neq 0$ 非类光性 2) 静态, Birkhoff 定理: 仅静电场

2. 轴对称解:

g_{ab} 为轴对称的 $\rightarrow$ $\exists$ 单参等度群, 轨道闭合类空 $\Leftrightarrow \exists$ 类空 Killing 场 ψ^a , 曲线闭合
axisymmetric

稳态: 类时 Killing ξ^a $[\xi, \psi]^a = 0 \rightarrow$ 坐标系 $\xi^a = (\partial/\partial t)^a$, $\psi^a = (\partial/\partial \varphi)^a$. $\partial g_{\mu\nu}/\partial t = (\mathcal{L}_\xi g)_{\mu\nu} = 0$, $\partial g_{\mu\nu}/\partial \varphi = (\mathcal{L}_\psi g)_{\mu\nu} = 0$

$\downarrow \forall p \in M, \exists$ 面 S 过 p , $S \perp \xi|_p, S \perp \psi|_p$ (S 上矢量 $u^a \xi_a = 0|_p, u^a \psi_a = 0|_p$) $\rightarrow$ 取 2 分量 x^2, x^3 , 沿 ξ^a 与 ψ^a 积分曲线不变

$[x_0=t, x_1=\varphi, x_2, x_3]$: $g_{\delta i} = 0$ ($\delta=0,1; i=2,3$), 令 $V \equiv -g_{00}, W \equiv g_{01}, X \equiv g_{11}$,

$$ds^2 = -V dt^2 + X d\varphi^2 + 2W dt d\varphi + g_{22}(dx^2)^2 + g_{33}(dx^3)^2 + 2g_{23} dx^2 dx^3$$

$\rightarrow$ 推 x_2, x_3 : 定义 $P^i \equiv Vx^i + W^i \rightarrow \frac{\partial P^i}{\partial t} = \frac{\partial P^i}{\partial \varphi} = 0 \rightarrow \xi^a \nabla_a P = 0, \psi^a \nabla_a P = 0 \rightarrow$ 以 P 为 $x_2 \rightarrow$
(满足 $g_{23}|_S = 0$: $\xi^a \nabla_a g_{23} = \mathcal{L}_\xi (g_{ab} \psi^a x^i)^\flat = g_{ab} \mathcal{L}_\xi (\psi^a x^i)^\flat + g_{ab} (x^i)^\flat \mathcal{L}_\xi (\psi^a)^\flat = g_{ab} [\underbrace{[\xi, \psi]^a}_{=0} x^i + \psi^a x^i]^\flat + c.c. = 0$)
故 $g_{23} = 0$ 对 S 外有效

- 1) 取等 P 值任定义 z
- 2) ∇P 积分曲线携带 z
- 3) ξ^a 与 ψ^a 携带 r, z

使 $\Omega^2 \equiv g_{22}, \Lambda \equiv g_{33}/\Omega^2, g_{23} = 0, w = W/V$

$$ds^2 = -V(dt^2 - w d\varphi^2) + V^{-1} P^2 d\varphi^2 + \Omega^2 (dp^2 + \Lambda dz^2)$$

3. 平面对称度规 (不是镜面对称)

对2维平面, killing场有3 ($E(2)$)

平面对称度规: g_{ab} 等度规群含子群 G_3 与 $E(2)$ 同构, G_3 的轨道都为2维平面
plane symmetric

Taub证明: $ds^2 = \frac{1}{\sqrt{1+\kappa Z}} (-dT^2 + dZ^2) + (1+\kappa Z)(dx^2 + dy^2)$

↓ 对称性: $ds^2 = \frac{1}{\sqrt{1+\kappa Z}} (-dT^2 + dZ^2) + (1+\kappa Z)(dx^2 + dy^2)$ 也可

↓ $\kappa \neq 0$ 时
 $t = \kappa^{-\frac{1}{3}} T, z = \kappa^{-\frac{2}{3}} (1+\kappa Z), x = \kappa^{\frac{1}{3}} X, y = \kappa^{\frac{1}{3}} Y$

$ds^2 = dz^{\frac{3}{2}} (-dt^2 + dz^2) + Z(dx^2 + dy^2)$ $\kappa \neq 0$ 时代表相同度规

4. Newman-Penrose 形式

类光标架法：从正交归一标架 $\{e_i\}$ 导出：

$$m^a := \frac{1}{\sqrt{2}}(e_1^a + i e_2^a), \quad \bar{m}^a := \frac{1}{\sqrt{2}}(e_1^a - i e_2^a)$$

$$l^a := \frac{1}{\sqrt{2}}(e_0^a + e_3^a), \quad k^a := \frac{1}{\sqrt{2}}(e_0^a - e_3^a)$$

$g_{ab} m^a m^b = 0$, 其余相同, 类光矢量, 记为 $(\varepsilon_i)^a_{(1,2,3,4)}$, 对应 $(\varepsilon^1)_a \equiv \bar{m}_a, (\varepsilon^2)_a \equiv m_a; (\varepsilon^3)_a \equiv -k_a, (\varepsilon^4)_a \equiv l_a$

分程 $g_{\mu\nu} = \begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 0 & -1 \\ 0 & 0 \end{bmatrix} = g^{\mu\nu} \rightarrow$ 刚性标架

里奇旋转系数 $w_{\mu\nu\rho} = (e_\mu)^a (e_\nu)^b \nabla_a (e_\rho)_b$: Claim: $w_{\mu\nu\rho}$ 中 1,2 互换得到共轭项 Pf: $\bar{w}_{\mu\nu\rho} = (\bar{e}_\mu)^a (\bar{e}_\nu)^b \nabla_a (\bar{e}_\rho)_b$, 易得.

命题对数量也有效: $\bar{w}_n = \overline{w_n}, \bar{w}_{12} = \overline{w_{21}}, \bar{R}_{12} = \overline{R_{21}} \dots$

嘉当结构方程: $\bar{R}_{\mu\nu} = d\bar{w}_{\mu\nu} + \bar{w}_\mu^\tau \wedge \bar{w}_{\tau\nu} = d\bar{w}_{\mu\nu} + g^{\lambda\tau} \bar{w}_{\mu\lambda} \wedge \bar{w}_{\tau\nu} \rightarrow$

$$\left[\begin{array}{l} R_{43} = d\bar{w}_{43} + \bar{w}_{32} \wedge \bar{w}_{41} - \overline{\bar{w}_{32} \wedge \bar{w}_{41}} = d\bar{w}_{43} + 2\text{Re}(\bar{w}_{32} \wedge \bar{w}_{41}) \\ R_{12} = d\bar{w}_{43} + 2i \cdot \text{Im}(\bar{w}_{32} \wedge \bar{w}_{41}) \end{array} \right. \quad \left[\begin{array}{l} \bar{R}_{43} = d\bar{w}_{43} + \bar{w}_{41} \wedge \bar{w}_{32} + \bar{w}_{42} \wedge \bar{w}_{13} \\ \bar{R}_{42} = d\bar{w}_{42} + \bar{w}_{41} \wedge (\bar{w}_{12} + \bar{w}_{43}), \bar{R}_{41} = \overline{\bar{R}_{42}} \\ \bar{R}_{32} = d\bar{w}_{32} + \bar{w}_{32} \wedge (\bar{w}_{12} + \bar{w}_{34}), \bar{R}_{31} = \overline{\bar{R}_{32}} \\ \bar{R}_{21} = d\bar{w}_{21} - \bar{w}_{13} \wedge \bar{w}_{42} - \bar{w}_{41} \wedge \bar{w}_{32} \end{array} \right.$$

$$\left[\begin{array}{l} \bar{R}_{42} = d\bar{w}_{42} + \bar{w}_{41} \wedge (\bar{w}_{12} + \bar{w}_{43}) \\ \bar{R}_{32} = d\bar{w}_{32} + \bar{w}_{32} \wedge (\bar{w}_{12} + \bar{w}_{34}) \\ \bar{R}_{21} + \bar{R}_{43} = d(\bar{w}_{21} + \bar{w}_{43}) + \bar{w}_{32} \wedge \bar{w}_{41} \end{array} \right]$$